sustainability

MDPI

Article

Domain-Specific Question-Answering Systems: A Case Study of a Carbon Neutrality Knowledge Base

Lei Liu [1,2,3], Yongzhang Zhou [1,2,3,*], Jianhua Ma [1,2,3], Yuqing Zhang [1,2,3] and Luhao He [1,2,3,*]

[1] Center for Earth Environment & Resources, Sun Yat-sen University, Zhuhai 519000, China; liulei58@mail2.sysu.edu.cn (L.L.); 05161935@cumt.edu.cn (J.M.); zhangyq363@mail2.sysu.edu.cn (Y.Z.)
[2] School of Earth Sciences and Engineering, Sun Yat-sen University, Zhuhai 519000, China
[3] Guangdong Provincial Key Laboratory of Mineral Resources and Geological Processes, Center of Earth Environment & Resources, Sun Yat-sen University, Zhuhai 519000, China
* Correspondence: zhouyz@mail.sysu.edu.cn (Y.Z.); helh8@mail2.sysu.edu.cn (L.H.)

Abstract: Carbon neutrality is a critical global objective in the fight against climate change. As relevant knowledge and technologies advance rapidly, there is an escalating demand for sophisticated intelligent services. While large language models (LLMs) have demonstrated considerable promise in knowledge processing and generation, their application within the domain of carbon neutrality remains in the early stages of exploration. This study develops a carbon neutrality knowledge base (CN Knowledge Base) using the ChatGLM3 model aimed at enhancing question-answering capabilities in areas such as carbon emission monitoring, policy interpretation, and technical optimization. By refining domain-specific corpora and integrating a Retrieval-Augmented Generation (RAG) mechanism, the model's ability to generate accurate and relevant responses is improved. To evaluate the performance of the proposed system, a comprehensive quantitative comparison is conducted using established evaluation metrics, including BLEU (Bilingual Evaluation Understudy), BERT (Bidirectional Encoder Representations from Transformers), and METEOR (Metric for Evaluation of Translation with Explicit Ordering). The CN Knowledge Base is benchmarked against leading models such as GPT-4, Gemini, and Bing. The results demonstrate that the CN Knowledge Base outperforms other models in METEOR (0.2697) and is comparable to GPT-4o in both BLEU (0.8755) and BERT (0.8260) Scores (GPT-4o: BLEU: 0.8784, BERT: 0.8404). These findings underscore the knowledge base's strong adaptability and its ability to generate high-quality, domain-specific content. The study suggests that specialized models can overcome the limitations of general-purpose LLMs, particularly in precise terminology and accurate application of domain knowledge. With continued development, such models could significantly enhance digital and intelligent solutions for carbon neutrality and related fields.

Keywords: domain-specific question answering; knowledge base construction; ChatGLM3; carbon neutrality; RAG

check for updates

Academic Editor: Hyunchul Ahn

Received: 2 January 2025
Revised: 28 February 2025
Accepted: 28 February 2025
Published: 3 March 2025

Citation: Liu, L.; Zhou, Y.; Ma, J.; Zhang, Y.; He, L. Domain-Specific Question-Answering Systems: A Case Study of a Carbon Neutrality Knowledge Base. *Sustainability* **2025**, *17*, 2192. https://doi.org/10.3390/su17052192

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Human activities have markedly increased the concentration of greenhouse gases (GHGs) in the global atmosphere, resulting in significant alterations to the climate system. To date, the global average surface temperature has risen by 1.1 °C compared to pre-industrial levels, with climate change increasingly threatening both ecosystems and human societies [1–3]. In response, countries worldwide have pledged to achieve carbon neutrality by the middle of the century, with the overarching goal of limiting the global temperature

rise to well below 1.5 °C. As a central strategy in the fight against climate change, carbon neutrality encompasses critical aspects such as energy system transformation, technological innovation, and policy development. However, nations encounter diverse challenges in advancing this ambitious objective [4–6]. Consequently, in pursuit of carbon neutrality, countries must not only transform their energy systems and foster technological breakthroughs, but also develop context-specific strategies to address the unique challenges they face [7,8]. Coordinating efforts across these domains and developing comprehensive solutions have emerged as critical challenges [9].

Achieving carbon neutrality is a multifaceted, complex endeavor that requires extensive interdisciplinary collaboration. Technological advances within a single domain are insufficient to address the full breadth of the challenges involved [10,11]. The body of research in this field spans a wide array of disciplines, including environmental monitoring, policy analysis, and carbon emission measurement [12,13]. Some of the primary challenges are the integration of heterogeneous data sources and the extraction of meaningful insights [14–16]. The effective and precise integration of diverse data and the extraction of key knowledge have become essential for advancing carbon neutrality research.

With the rapid development of artificial intelligence technologies, large language models (LLMs) have made significant strides in the field of natural language processing (NLP) [17], providing strong support for carbon neutrality research. Thanks to their advanced semantic understanding and reasoning capabilities, LLMs have become instrumental in areas such as climate change forecasting, carbon emission monitoring, and the formulation of sustainable development strategies [18–20]. By integrating domain-specific knowledge, LLMs are able to efficiently synthesize climate data, scientific findings, and policy information, thereby offering precise analytical support and decision-making guidance to achieve carbon neutrality objectives [21]. However, general-purpose LLMs, which are typically trained on broad, generic datasets, often face limitations when addressing the specific challenges of specialized domains [22]. In the absence of domain-specific expertise, the responses generated by these models can lack both precision and relevance [23]. The development of domain-specific question-answering (DSQA) systems has emerged as a critical strategy to enhance the performance of large language models in specialized fields.

Question-answering (QA) systems are a vital application within the field of NLP, designed to provide accurate answers to specific user queries [24]. These systems are generally categorized into "closed-domain" (CDQA) and "open-domain" (ODQA) systems, depending on the knowledge domain. Closed-domain systems focus on answering domain-specific questions, while open-domain systems address a broader range of topics [25]. However, traditional QA systems, including those based on LLMs, are often insufficiently trained for specialized domains, making it difficult for them to understand domain-specific knowledge and terminology. To enhance their performance for particular tasks and domains, these models generally require fine-tuning [26,27]. For instance, Vulpescu et al. (2024) [28] fine-tuned the Llama model to improve its performance across various domains, achieving high accuracy in topic classification. Lee et al. (2024) [29] enhanced models such as ChatGPT3 using prompt engineering techniques, developing an Automatic Question Generation (AQG) system for English education. Giray (2023) [30] explored how prompt engineering can improve the output quality of ChatGPT in academic writing. While prompt engineering and fine-tuning can boost performance in specialized domains, overfitting to a particular domain may lead to hallucinations, resulting in erroneous outputs [31]. To mitigate this issue and improve generalization, Retrieval-Augmented Generation (RAG) technology has gained significant attention [32]. This approach integrates pre-trained models with external knowledge bases, enabling the effective extraction of domain-specific information and the generation of accurate, contextually relevant responses [33].

In recent years, knowledge-graph-based methods and RAG with LLMs have demonstrated substantial advantages in DSQA systems [34]. Numerous studies have utilized knowledge graphs to develop and enhance QA systems [35,36]. For example, Sovrano et al. (2020) [37], Bulla et al. (2021) [38], and Liu et al. (2023) [39] created QA systems for the legal, financial, and mining sectors, respectively, by leveraging knowledge graphs. These systems have demonstrated the ability to effectively comprehend questions, perform reasoning, and generate contextually relevant and accurate answers [40]. In addition, Jia et al. (2021) [41] developed an end-to-end QA system using knowledge graphs to address complex temporal questions involving multiple entities, predicates, and time conditions. Yang et al. (2021) [42] combined knowledge graphs with intelligent QA technologies to design a smart system for high school education. Despite these advancements, knowledge-graph-based systems still face challenges related to semantic understanding and knowledge updating [43]. To further enhance QA system performance, Wang et al. (2024) [44], Soman et al. (2024) [45], Oduro-Afriyie (2024) [46], and Bui et al. (2024) [47] applied technology, based on knowledge graphs, to develop DSQA systems in fields such as medicine and education. These systems have demonstrated efficient retrieval of domain-specific knowledge and have significantly enhanced user satisfaction through system optimization and training.

While intelligent QA systems have made significant progress in various domains, research in the field of carbon neutrality remains at an early stage. Existing carbon neutrality QA systems have not yet effectively integrated cross-disciplinary, multi-source knowledge, which limits their capacity to provide accurate solutions to the complex challenges inherent in carbon neutrality practices. Therefore, developing an intelligent QA system that synthesizes knowledge from diverse fields, including environmental science, energy policy, climate forecasting, and carbon emission technologies, has become a pressing challenge.

This paper proposes a domain-specific QA system for carbon neutrality based on RAG with an external knowledge base and the ChatGLM3-6B model integrated with the Langchain open-source framework. The system combines expertise from environmental science, energy policy, climate forecasting, and carbon emission technologies. It automates data processing and delivers precise analyses and recommendations, enhancing the system's ability to address questions related to carbon neutrality. The research steps are outlined as follows: (1) system design and implementation using the ChatGLM3-6B model integrated with the Langchain framework; (2) construction of a carbon neutrality domain knowledge base by collecting relevant academic papers, reports, and books; (3) optimization of system performance through rigorous evaluation and testing. This study represents the first in-depth investigation into the application of the ChatGLM3-based large language model for constructing a carbon neutrality knowledge base (hereinafter referred to as the CN Knowledge Base) and an intelligent QA system. It highlights the distinct advantages of domain-specific large language models in addressing specialized tasks. By developing a knowledge base specifically focused on carbon neutrality, this research not only demonstrates the superior efficiency and accuracy of domain-specific models over general-purpose models in specialized domains, but also introduces a novel framework that provides powerful, intelligent tools to support carbon neutrality research and inform policy decision-making. This work contributes to the advancement of carbon neutrality technologies and offers critical technical support and decision-making assistance in achieving the global goal of carbon neutrality.

2. Method

2.1. ChatGLM3-6B Model

ChatGLM is a next-generation, conversational, pre-trained model co-developed by Zhipu AI and the KEG Lab at Tsinghua University [48]. This model is based on the autoregressive, blank-filling language model (GLM) introduced by Du et al. (2021) [49], which outperforms BERT on the SuperGLUE benchmark and significantly exceeds the performance of T5 and GPT in both natural language understanding (NLU) and generation tasks. ChatGLM3-6B, an optimized bilingual conversational model, has specifically enhanced capabilities for processing Chinese. It employs semantic unit tokenization and Rotary Position Encoding (RoPE), which significantly improve its ability to understand and generate long texts. Additionally, the model supports over 20 languages, making it highly effective for multilingual tasks.

In addition, ChatGLM3-6B exhibits strong reasoning capabilities, leveraging quantization techniques to enhance computational efficiency. It also supports lightweight fine-tuning, making it suitable for deployment on resource-constrained devices. When combined with RAG and security filtering mechanisms, ChatGLM3-6B demonstrates exceptional performance in knowledge-based QA and real-time information processing while ensuring the safety and compliance of the generated content. The model excels in tasks such as text generation, reasoning, and QA, efficiently handling long-text generation, multi-turn dialogues, and complex reasoning tasks. It has achieved top performance in 42 benchmark tests, covering areas such as semantic understanding, mathematical reasoning, inferential reasoning, programming, and knowledge processing. This makes it particularly well suited for both ODQA and domain-specific tasks [50].

2.2. Langchain Framework

LangChain is an open-source framework designed to build intelligent applications powered by LLMs. By integrating external data sources, tools, and business logic, LangChain significantly enhances the capabilities of language models, enabling them to perform tasks such as automation, context management, and dynamic response generation. This broadens its applicability to complex domains such as intelligent QA, document processing, and knowledge base construction. LangChain's core goal is to provide developers with a modular, flexible, and scalable toolkit for creating more advanced and interactive applications [51]. The Chinese knowledge base QA application LangChain-Chatchat, developed using the LangChain framework, combines LLMs such as ChatGLM to offer an open-source, offline-deployable RAG system. This system efficiently integrates LLMs with external knowledge bases, delivering accurate knowledge-based QA and information-processing services. It highlights the significant advantages of LangChain in enabling the effective application of LLMs [52].

3. Construction and Evaluation Methods of the Question-Answering Framework

3.1. Architecture of the Question-Answering System

The architecture of the proposed QA system is show in Figure 1, and consists of three core modules: Local Documents (Document Repository), Query, and Answer. Initially, local documents are converted into text chunks, which are then embedded into vector representations and stored in a vector database. Upon receiving a user query, the system transforms the query into a vector and compares it for similarity with the stored document vectors to retrieve the most relevant text chunks. Subsequently, the system employs prompt templates in conjunction with LLMs to generate and deliver the final response. This approach effectively integrates text processing, vector storage, and LLM-based inference,

ensuring accurate and precise answers to user queries. In this framework, ChatGLM3-6B serves as the core LLMs responsible for natural language processing tasks. By integrating LangChain with ChatGLM3-6B and utilizing RAG technology, the system retrieves relevant information from the local document repository, refines it, and generates coherent responses. This method ensures the delivery of high-quality, contextually relevant answers.

3.1.1. Local Documents

The core function of this module is to construct the knowledge base by parsing local documents and converting them into structured data that can be efficiently retrieved and matched. The processed data are then stored in a vector database, ensuring that the system is equipped with a robust knowledge base for subsequent queries and answer generation. The specific process is as follows: Local Documents → Unstructured Loader → Text → Text Splitter → Text Chunks → Embedding → VectorStore.

Firstly, the system loads local documents using the Unstructured Loader, extracting unstructured content into text data. In the Text Preprocessing step, the extracted text is processed to ensure it is suitable for further operations, such as cleaning and standardization. The Text Splitter then divides lengthy documents into smaller segments to improve the efficiency and accuracy of information retrieval. By partitioning documents into smaller units, the system can more precisely locate relevant content. In the Text Chunks step, the split text segments are further organized to enhance their usability during the vectorization process. Secondly, in the Embedding step, the text segments are passed to an embedding model, where they are converted into high-dimensional vectors. This process captures the semantic information of the text, allowing for more accurate understanding and matching during subsequent vector searches. Finally, the transformed vectors are stored in the VectorStore, forming the foundation for future queries and similarity searches. During a query, the system efficiently retrieves the most relevant knowledge fragments from the VectorStore, enabling precise answer generation. This series of processes ensures that the Local Documents module can efficiently handle and store large volumes of textual data, providing a solid foundation for knowledge retrieval and answer generation within the QA system.

3.1.2. Query

The core function of this module is to receive user queries and retrieve relevant text segments. By transforming the user's query into an embedding vector and performing similarity computations with the vector representations stored in the vector database, the system can efficiently match the most relevant text segments to support answer generation. The process is as follows: Query → Embedding → Query Vector → Vector Similarity → Related Text Chunks.

First, the system receives the user's input query and transforms it into an embedding vector during the Embedding step. This converts the query into a high-dimensional vector, enabling efficient similarity comparisons with the stored content in the vector database. The Query Vector represents the user query as a vector, which is then matched with the stored embedding vectors to ensure the query accurately reflects the user's intent. In the subsequent Similarity Calculation step, the system computes the similarity between the query vector and the database vectors, quickly identifying the text segments most relevant to the query. This similarity computation relies on distance measures in the vector space (e.g., cosine similarity), allowing the system to efficiently identify the most relevant content. Finally, the system returns the Relevant Text Segments, which serve as the basis for generating the answer. Through these steps, the Query module ensures that the system can swiftly retrieve the most relevant information from a vast knowledge base, supporting the generation of precise and contextually relevant answers.

3.1.3. Answer

The primary function of this module is to generate answers. By utilizing the relevant text fragments retrieved from the database alongside the user query, the system produces natural language responses using the LLM. The process is as follows: Related Text Chunks → Prompt Template → Prompt → LLM → Answer.

First, the system retrieves the relevant text fragments from the Related Text Chunks, which provide the necessary contextual information for generating the answer. Next, the relevant text fragments and the user query are structured using the Prompt Template, creating a format suitable for processing by the language model. This prompt is then passed to the model to generate the final response. The LLM receives the prompt and, leveraging its language understanding and generation capabilities, transforms the retrieved knowledge into fluent, natural language. Finally, the system generates the answer as a response to the user query.

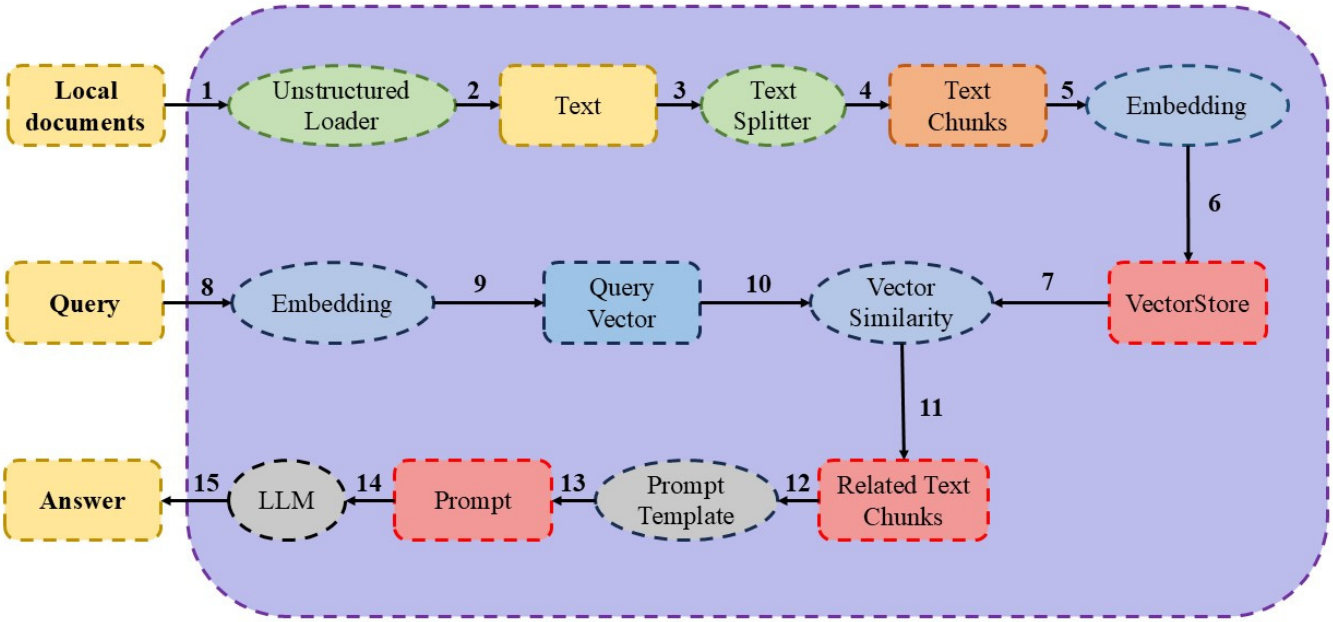


Figure 1. Question-answering system framework.

3.2. Dataset Construction

The CN Knowledge Base constructed in this study comprises 6003 research papers sourced from prominent academic platforms such as Web of Science, Google Scholar, and CNKI (China National Knowledge Infrastructure). It comprehensively covers a broad spectrum of research findings and the latest advancements in technology, policy, economics, and environmental studies, all pertinent to the field of carbon neutrality. The selection of the literature was based on criteria such as academic authority, update frequency, and its overall contribution to the domain, ensuring that both core knowledge and cutting-edge developments are represented. The collected documents were preprocessed to generate clean text data, eliminating noise and redundant information to improve the accuracy of subsequent processing. After preprocessing, the documents were divided into smaller text chunks, each capturing the essential knowledge of the source. These chunks were then passed to ChatGLM3 for vectorization, transforming the content into high-dimensional vectors that encapsulate the deep semantic meaning of the text. These vectors preserve not only the syntactic structure but also the underlying concepts, enhancing the precision of future retrieval tasks.

The vectorized text segments are stored in a vector database, which employs an efficient indexing mechanism to enable rapid retrieval of relevant text segments in response

to user queries. Upon receiving a query, the system converts it into an embedding vector. By calculating the similarity between the query vector and the stored vectors, the system identifies and returns the most relevant text segments. This process ensures highly accurate and efficient information retrieval, minimizing redundancy and noise, and enables the system to deliver precise, contextually relevant knowledge tailored to the user's needs.

3.3. Evaluation Method

To systematically evaluate the CN Knowledge Base, this study will establish an evaluation framework that includes representative questions and answers from the carbon neutrality domain. The selection of 500 questions carefully considers the diversity and complexity of the field, ensuring coverage across multiple subdomains, including technology, policy, economics, and environmental studies. The questions will span a range of difficulty levels, from basic to advanced, allowing for an evaluation of the system's performance in handling questions of varying technical depth and complexity. Additionally, the questions will reflect the latest trends and emerging research in the carbon neutrality field, ensuring that the system can address current industry challenges and practical needs. The standard answer repository will primarily consist of authoritative sources, such as academic journals and textbooks, to ensure the reliability and accuracy of the responses, with sample questions and answers provided in Table 1.

The evaluation will involve eight models: the ChatGLM3-6B base model, the CN Knowledge Base, ChatGPT-4 (GPT-4o) [53], Gemini [54], Bing [55], Kimi [56], Doubao [57], and ERNIE Bot [58]. Each model will generate 500 answers, which will then be evaluated against the predefined question set. The answers will be quantitatively assessed using standard NLP metrics, including BLEU [59], BERT Score [60], and METEOR [61]. The evaluation will cover multiple dimensions, such as semantic accuracy, fluency, completeness, and similarity to the standard answers. This comprehensive evaluation provides robust data to optimize the CN Knowledge Base and offers valuable insights for the future development and enhancement of intelligent QA systems.

3.4. Evaluation Algorithms

3.4.1. BLEU

BLEU (Bilingual Evaluation Understudy) is a widely used automated metric in NLP for evaluating the quality of generated text. It calculates the degree of overlap between n-grams (i.e., contiguous sequences of n words) in the generated text and a reference text. The fundamental principle behind BLEU is that the higher the n-gram similarity between the generated text and the reference, the higher the quality of the generated output. BLEU Scores are typically computed using different n-gram sizes (e.g., unigram, bigram, trigram) and are often weighted to prioritize higher-order n-grams. F1 Score is another widely employed evaluation metric in NLP that combines two important concepts: Precision and Recall. Precision (1) measures the proportion of relevant items retrieved from the total items retrieved, ensuring that the generated output is accurate and relevant. Recall (2) measures the proportion of relevant items retrieved from the total relevant items available, ensuring that the generated output is complete and covers the necessary content. The F1 Score (3) provides a harmonic mean of Precision and Recall, offering a balanced evaluation metric. It is particularly useful in cases where you want to balance between the two metrics. By combining them, the F1 Score helps avoid the limitations of using only Precision (which might lead to overly short or incomplete responses) or only Recall (which might produce redundant or overly long responses). This combined approach ensures a more comprehensive and fair assessment of the generated text [59].

$$Precision = \frac{Number\ of\ matched\ n-grams}{Total\ number\ of\ n-grams\ generated} \quad (1)$$

$$Recall = \frac{Number\ of\ matched\ n-grams}{Total\ number\ of\ n-grams\ in\ reference} \quad (2)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

Table 1. Professional question and answer standard library.

Question	Reference Answer
What role does carbon capture and storage (CCS) technology play in carbon neutrality?	Carbon capture and storage (CCS) technology plays a crucial role in achieving carbon neutrality. It involves three primary steps: capture, transportation, and storage. First, CO_2 is captured from industrial sources (such as coal-fired power plants, cement factories, and steel mills) through chemical or physical methods. The captured CO_2 is then transported via pipelines, ships, or trucks to designated storage sites. Finally, it is injected into geological formations—such as depleted oil and gas fields, deep saline aquifers, or coal seams—for long-term sequestration, preventing its release into the atmosphere. CCS contributes to carbon neutrality by: Directly reducing industrial emissions: Many heavy industries and the energy sector face challenges in achieving deep emissions reductions by other means. CCS can directly capture their CO_2 emissions, significantly mitigating their environmental impact, which is essential for meeting short-term emission reduction targets. Supporting energy transition: During the transition to cleaner energy, CCS helps existing fossil-fuel-based power plants reduce emissions gradually, extending their operational lifespan while facilitating the integration of renewable energy sources. For example, natural gas power generation combined with CCS can act as a transitional energy source, reducing the carbon footprint of the power sector. Enabling negative emissions technologies: CCS can be integrated with biomass energy to create Bioenergy with Carbon Capture and Storage (BECCS). BECCS involves growing biomass crops that absorb CO_2, which is then captured and stored during combustion, enabling negative emissions and further reducing atmospheric CO_2. Economic feasibility: While CCS is currently costly, technological advancements and economies of scale are expected to drive down costs. Additionally, CCS has been successfully demonstrated in several countries, proving its feasibility and reliability. Policy support and international cooperation: The development and widespread adoption of CCS requires strong policy support and international collaboration. Governments can incentivize investment in CCS through financial subsidies, tax incentives, and legislative measures. Global cooperation is also vital for sharing technological knowledge and accelerating the deployment of CCS worldwide.
What is a “low-carbon lifestyle”, and what is its significance for carbon neutrality?	A low-carbon lifestyle refers to a way of living that minimizes carbon emissions, primarily from CO_2, by adopting energy-efficient practices, reducing consumption, and lowering overall environmental impact. This movement has gained traction in major cities across China and is subtly reshaping people’s daily habits. A low-carbon lifestyle is characterized by healthier, more sustainable choices that harmonize human activities with nature. As living standards have improved, so have their environmental impacts. A low-carbon lifestyle not only represents a shift in personal habits but also embodies a commitment to sustainable environmental practices. For carbon neutrality, a low-carbon lifestyle plays a critical role. By reducing individual carbon footprints through actions like using energy-efficient appliances, choosing public transportation, and consuming less meat, everyone can contribute to lowering greenhouse gas emissions. Moreover, when more people adopt such practices, the cumulative effect can lead to significant reductions in overall emissions. Additionally, this lifestyle encourages businesses and markets to adopt greener practices and raises public awareness of environmental issues, fostering a collective effort to meet carbon neutrality goals.
How do remote sensing and machine learning technologies contribute to real-time monitoring of forest carbon stock changes?	Remote Sensing Technology: Remote sensing enables the collection of multi-source data about forests using various sensors, such as optical satellite images, radar data, hyperspectral images, and LiDAR (Light Detection and Ranging). These data provide valuable information on forest structure, biomass, and health. For instance, satellite imagery reveals forest cover and vegetation indices, radar data penetrates cloud cover to provide surface details, hyperspectral images capture vegetation spectral characteristics, and LiDAR generates high-precision 3D models of terrain and vegetation. By analyzing time-series data, changes in forests can be monitored over time, including growth, deforestation, and fire events. For example, examining the variations in NDVI (Normalized Difference Vegetation Index) over time helps assess forest growth and carbon stock changes. Machine Learning Technology: Machine learning techniques can automatically extract key features from remote sensing data, such as vegetation indices, texture, and spectral features. These features are then used in classification and regression models to identify different vegetation types and estimate carbon stocks. Convolutional Neural Networks (CNNs) can be used to extract spectral features from hyperspectral images, while Support Vector Machines (SVMs) are useful for classifying vegetation types. Machine learning models can be trained using historical data and field measurements to predict changes in forest carbon stocks. Common algorithms include random forests, gradient boosting trees (GBDT), support vector regression (SVR), and deep learning models like Long Short-Term Memory (LSTM). By combining remote sensing data with machine learning models, real-time monitoring and early warnings of carbon stock changes can be achieved. For example, satellite and drone imagery can provide real-time data on forest health, and machine learning models can quickly assess changes in carbon stocks. If anomalies are detected, alerts can be generated, triggering appropriate management actions. These technologies provide efficient, accurate data for real-time monitoring of forest carbon stocks, essential for forest management and carbon reduction efforts.

3.4.2. BERT Score

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model based on the Transformer architecture. It learns rich contextual representations by pre-training on large-scale text data, significantly enhancing natural language understanding. BERT's effectiveness has led to its widespread application in various NLP tasks, such as text classification, named-entity recognition (NER), QA, and text generation. Unlike evaluation methods such as BLEU or F1 Score, which focus on the surface-level matching of n-grams, BERT Score evaluates the quality of generated text based on its semantic understanding. Instead of simply matching word sequences, it assesses the similarity of words at a deeper level, considering their meaning in context. The process of calculating the BERT Score is as follows: (1) Text Encoding: Both the generated text and the reference text are encoded into contextualized word vectors (embeddings) using the BERT model. These word vectors capture the semantic meaning of each word in its context, leveraging BERT's bidirectional Transformer architecture. (2) Word Vector Similarity Calculation: For each word w_i in the generated text, the similarity with all words r_j in the reference text is calculated. This step measures how closely each word in the generated text aligns semantically with the words in the reference text (Equation (4)). (3) For each word in the generated text, the system matches it to all word vectors in the reference text, selecting the word vector with the highest similarity as its corresponding reference word (Equation (5)). (4) Matching Score Calculation: The maximum similarity score between each word in the generated text and its most similar word in the reference text is computed. Finally, the average of these similarity scores is taken to yield the BERT Score, as shown in (6) [60].

$$sim(w_i, r_j) = \frac{V_i \times V_j}{\|V_i\| \times \|V_j\|} \tag{4}$$

$$\underset{r_j \in R}{max}\ sim(w_i, r_j) \tag{5}$$

$$BERT\ Score = \frac{1}{|G|} \times \sum_{i=1}^{|G|} \underset{r_j \in R}{max}\ sim(w_i, r_j) \tag{6}$$

where V_i, V_j are the BERT embedding vectors corresponding to the words w_i in the generated text and r_j in the reference text, and $|G|$ is the total number of words in the generated text.

3.4.3. METEOR

METEOR (Metric for Evaluation of Translation with Explicit Ordering) is a metric developed to assess the quality of automatically generated text by comparing it with reference text. Unlike traditional n-gram-based evaluation methods like BLEU, which focus primarily on surface-level matching, METEOR adopts a more holistic approach. It incorporates various factors such as Precision, Recall, Synonym Matching, Morphological Variations, and Word Order Penalties, providing a more comprehensive and nuanced evaluation of the generated text's quality. The key principles behind METEOR are as follows: (1) Precision and Recall: METEOR begins by calculating the Precision and Recall between the generated and reference texts. Precision measures the proportion of matching words in the generated text relative to the total number of words in it, while Recall evaluates the proportion of words in the reference text that are covered by the generated text. To balance these two metrics, METEOR uses a weighted harmonic mean (7). (2) Synonym Matching: In contrast to BLEU, METEOR considers not only exact word matches but also synonym matches. It uses a synonym dictionary (such as WordNet) to identify words in the generated text that are synonyms of words in the reference text, treating these synonyms as valid matches. (3) Word Order Penalty: METEOR applies a penalty for discrepancies in

word order. If the word order in the generated text differs from the reference text, a penalty is applied to the matched words. The method for calculating this penalty is outlined in (8). (4) Final METEOR Score: The final METEOR Score is calculated by combining the weighted harmonic mean of Precision and Recall with the Word Order Penalty. The formula for this final score is provided in (9) [61].

$$F_{BERT} = \frac{Precision \times Recall}{\alpha \times Precision + Recall} \tag{7}$$

$$Penalty = \beta \times \frac{Number\ of\ disordered\ matches}{Total\ Number\ of\ matches} \tag{8}$$

$$METEOR\ score = (1 - Penalty) \times F_{BERT} \tag{9}$$

where α is the balance coefficient between Precision and Recall, typically set to 0.9. The weighted harmonic mean uses α to balance Precision and Recall, ensuring that neither metric dominates excessively. This prevents extreme scenarios, such as cases with high Precision but low Recall (or vice versa), thereby ensuring a more balanced evaluation.

β is a constant, usually set to 0.5 or 0.6, that controls the severity of the penalty for differences in word order. The larger the deviation in word order between the generated text and the reference text, the higher the penalty applied. This penalty lowers the final METEOR Score, reflecting the impact of word order on the quality of the generated text.

4. Results

4.1. Chat Page

As shown in Figure 2, querying the CN Knowledge Base using the ChatGLM3-6B model involves several stages: data retrieval, query representation, semantic matching, and answer generation. The following steps outline how the model processes the input question and generates accurate and relevant answers: (1) User Query: The user asks, "What are 10 examples of how artificial intelligence and big data can be used to optimize carbon monitoring and management systems to improve the efficiency of carbon-neutral policy implementation?" The model is tasked with generating 10 examples discussing how AI and big data can optimize carbon monitoring and management systems, thereby enhancing the efficiency of carbon-neutral policy implementation. (2) Query Conversion to Embedding Vectors: The model begins by converting the user's question into an embedding vector which the system can process. This is performed using an embedding model that transforms the question into a high-dimensional vector, encoding both the semantic and contextual information. Key terms such as "artificial intelligence", "big data", "carbon monitoring", and "policy implementation" are used to generate this vector. This vector serves as a query representation, enabling the system to efficiently search the CN Knowledge Base for relevant content. (3) Knowledge Base Retrieval: Using the generated query vector, the system applies RAG technology to search for the most relevant documents or information snippets within the CN Knowledge Base. These snippets, sourced from structured or unstructured text data that have been preprocessed and stored in a vector database, are retrieved by calculating the similarity between the query vector and the document embedding vectors. This enables the system to quickly identify content related to topics such as artificial intelligence, big data, carbon monitoring, and policy implementation. (4) Selection and Integration of Text Snippets: Once the relevant text snippets are retrieved, the system filters and integrates them to ensure their high relevance to the user query. For example, the retrieved content may include sections on "how artificial intelligence optimizes carbon monitoring systems" or "how big data enhances carbon-neutral policy implementation efficiency". This step ensures the generated answer

meets the user's request for "10 examples" and captures the relationship between AI, big data, and carbon monitoring systems. (5) Answer Generation: After integrating the relevant text snippets, the system uses the ChatGLM3-6B model to generate a natural language response. The model takes into account the semantic content, context, and specific requirements of the user's query (e.g., generating 10 examples). The generated answer is clear, structured, and coherent. By processing the Prompt Template and Prompt, the system combines the retrieved text snippets with the query, producing a well-organized response. The final answer will present 10 concrete examples illustrating how AI and big data can optimize carbon monitoring. (6) Answer Source Display: To ensure transparency and credibility, the system displays the sources of each answer. Every generated response will include links to the relevant knowledge base entries or document snippets, showing the origin of the information. This allows users not only to view the answer but also to explore detailed information from the corresponding literature or research. (7) Answer Presentation: The final answer is presented to the user in natural language, ensuring clarity, accuracy, and comprehensiveness. It will include specific examples directly addressing how artificial intelligence and big data can optimize carbon monitoring and management systems, providing a well-rounded response to the query.

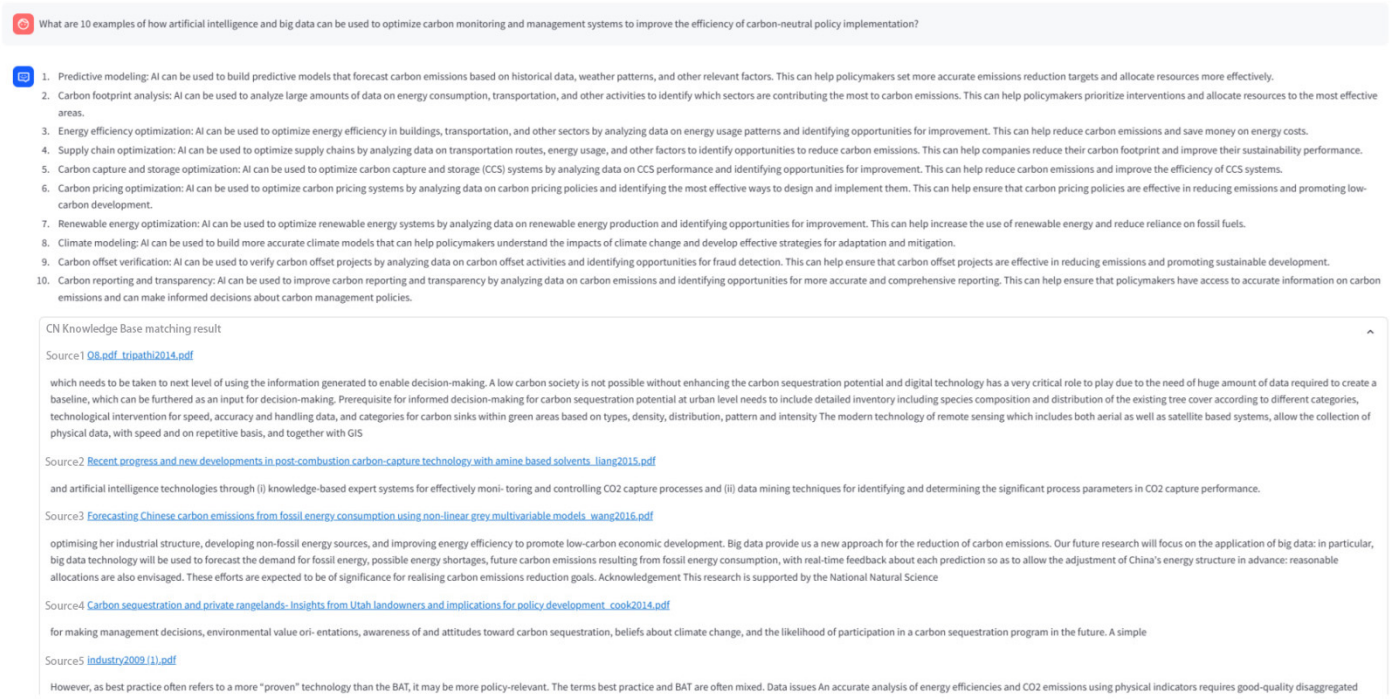


Figure 2. A dialog page based on the ChatGLM CN Knowledge Base.

4.2. Case Analysis

A comparative analysis of the CN Knowledge Base, GPT-4, Bing AI, and Gemini was conducted to assess their performance in answering questions related to carbon neutrality (Table 2). The evaluation focused on several key dimensions: content comprehensiveness, technical depth, practical applicability, and simplicity and comprehensibility.

Table 2. Answers from GPT-4, Bing AI, and Gemini.

	Answer
GPT-4	**1. Real-time Carbon Emission Monitoring:** AI can process data from sensors (e.g., IoT devices, satellites) to provide real-time tracking of CO_2 emissions from industries, transportation, and agriculture. This data, combined with big data analytics, allows governments and companies to identify emission hotspots and monitor compliance with emission reduction targets. **2. Predictive Modeling for Emission Reductions:** AI-powered machine learning models can predict future carbon emissions based on current trends and historical data. By analyzing large dataset. AI can forecast the impact of specific policies or technological interventions on emissions, helping policymakers make data-driven decisions. **3. Carbon Capture and Storage (CCS) Optimization:** AI can optimize the processes used in carbon capture, transport, and storage. Machine learning Algorithms can predict the efficiency of different ICS technologies, optimize energy use, and enhance decision-making around site selection for storage, reducing leakage risks and improving storage capacity. **4. Smart Grid and Energy Management:** AI can optimize the use of renewable energy in smart grids by predicting energy demand and supply, minimizing carbon emissions. Big data car De used to analyze energy consumption patterns helping to balance load and integrate renewable energy sources like solar and wind more effectively, thus lowering reliance on fossil fuels. **5. Carbon Footprint Estimation and Reduction for Enterprises:** AI Algorithms can analyze operational data from businesses to calculate their carbon footprint in real-time. This enables organizations to identify areas of high carbon intensity, optimize resource usage, and implement targeted strategies to reduce emissions, such as energy-efficient technologies or supply chain improvements. **6. Agricultural Carbon Management:** AI and big data can improve carbon sequestration in agriculture. AI can be used to analyze soil conditions, weather patterns, and crop types to recommend the best practices for soil carbon capture. Big data can help in monitoring agricultural emissions and improving and management to increase carbon storage. **7. Carbon Trading and Market Optimization:** AI can optimize carbon trading platforms by analyzing market data, price fluctuations, and regulatory changes, Machine learning models can predict carbon credit prices, identify trends, and improve decision-making for trading, ensuring a more efficient and transparent carbon market. **8. Carbon Footprint Reduction in Transportation:** AI algorithms can analyze traffic patterns, vehicle performance, and route optimization to reduce emissions from transportation. Big data from connected vehicles and transportation systems can be leveraged to improve fleet management, reduce fuel consumption, and transition to electric vehicles. **9. Carbon Accounting and Policy Impact Evaluation:** AI can automate the process of carbon accounting by integrating data from various sources, ensuring accurate tracking and reporting of emissions. It can also evaluate the effectiveness of carbon-neutral policies by analyzing data on emissions reductions, cost savings, and environmental impact helping governments refine their strategies. **10. Smart Cities and Urban Carbon Management:** AI and big data can be used in smart city initiatives to optimize energy use in buildings, public transportation, and waste management. AI algorithms can analyze urban carbon footprints, optimize infrastructure, and manage energy consumption, contributing to the overall goal of achieving carbon neutrality in cities.
Bing AI	**1. Predictive Analytics for Emissions:** AI can analyze historical emissions data to predict future trends and identify potential areas for improvement. **2. Satellite Imagery Analysis:** AI can process satellite images to monitor deforestation, track changes in land use, and measure carbon storage in forests. **3. Smart Grid Management:** AI can optimize energy distribution and reduce waste by predicting energy demand and integrating renewable energy sources. **4. Carbon Capture and Storage (CCS):** AI can help identify optimal locations for CCS facilities and improve the efficiency of carbon capture processes. **5. Precision Agriculture:** AI can analyze data from sensors and drones to optimize crop. management, reduce fertilizer use, and minimize greenhouse gas emissions from agriculture. **6. Traffic Management:** AI can optimize traffic flow in urban areas to reduce congestion and lower vehicle emissions. **7. Building Energy Management:** AI can monitor and control energy use in buildings, improving efficiency and reducing carbon footprints. **8. Supply Chain Optimization:** AI can analyze supply chain data to identify inefficiencies and reduce emissions from transportation and logistics. **9. Waste Management:** AI can optimize waste collection routes and processes to reduce emissions from waste management activities. **10. Public Awareness Campaigns:** AI can analyze social media and other data sources to tailor public awareness campaigns about carbon reduction and sustainability

Table 2. *Cont.*

	Answer
Gemini	**1. Real-time Emissions Monitoring and Forecasting:** **AI-powered sensor networks:** Deploy AI-driven sensor networks to continuously monitor emissions from various sources (e.g., power plants, factories, vehicles). **Predictive analytics:** Utilize historical data and machine learning to forecast future emissions trends, enabling proactive measures. **2. Optimized Carbon Capture, Utilization, and Storage (CCUS):** **Intelligent process control:** Employ AI to optimize CCUS processes. maximizing carbon capture efficiency and minimizing energy consumption. **Predictive maintenance:** Use AI to predict equipment failures in CCUS facilities, reducing downtime and improving overall system reliability. **3. Enhanced Renewable Energy Integration:** **Smart grid optimization:** Leverage AI to optimize energy distribution in smart grids, integrating renewable energy sources more efficiently. **Predictive energy forecasting:** Use AI to forecast renewable energy generation, enabling better grid management andenergy storage planning. **4. Improved Energy Efficiency in Buildings:** **AI-powered building automation:** Utilize AI to optimize heating, ventilation, and air conditioning (HVAC)systems reducing energy consumption. **Predictive maintenance for building infrastructure:** Employ AI to identify potential equipment failures in buildings, preventing energy losses. **5. Optimized Transportation and Logistics:** **Intelligent route planning:** Use AI to optimize delivery routes, reducing fuel consumption and emissions. **Real-time traffic management:** Employ AI to manage traffic flow, minimizing congestion and emissions. **6. Advanced Forest Monitoring and Management:** **Satellite imagery analysis:** Use AI to analyze satellite imagery to monitor deforestation and forest degradation. **Predictive forest fire modeling:** Utilize AI to predict the spread of forest fires, enabling early intervention and prevention. **7. Enhanced Carbon Accounting and Reporting:** **Automated data collection and analysis:** Employ AI to automate the collection and analysis of carbon data, improving accuracy and efficiency. **Blockchain-based carbon credit tracking:** Use blockchain technology to ensure the transparency and security of carbon credit transactions **8. Personalized Carbon Reduction Strategies:** **AI-powered carbon footprint calculators:** Develop personalized carbon footprint calculators to help individuals and or organizations identify areas for improvement. **Tailored carbon reduction plans:** Use AI to create customized carbon reduction plans based on individual or organizational needs and circumstances. **9. Climate Change impact Assessment and Adaptation:** **Climate modeling and simulation:** Utilize AI to simulate future climate scenarios and assess their potential impacts. **Climate adaptation planning:** Employ AI to develop effective climate adaptation strategies, such as coastal protection and infrastructure resilience. **10. International Collaboration and Data Sharing:** **AI-powered data integration and analysis:** Use AI to integrate and analyze data from various sources, facilitating international cooperation on climate change. **Global carbon monitoring platforms:** Develop global carbon monitoring platforms to track emissions and progress towards climate goals.

(1) Content Comprehensiveness: The CN Knowledge Base provides detailed, expansive responses, covering a wide range of topics in the carbon neutrality field, such as carbon emission forecasting, energy efficiency optimization, CCS technologies, and supply chain management. It offers both broad coverage and significant depth, making it highly valuable for policymakers and industry professionals. In contrast, GPT-4 also addresses essential topics like carbon monitoring, carbon capture technologies, and smart grid systems, but places a stronger emphasis on the applications of AI and big data in carbon neutrality. It does not delve as deeply into policy implementation, focusing more on technological solutions. Bing AI's responses are more concise and center around topics like carbon emission

prediction, satellite image analysis, and intelligent transportation systems. However, its coverage is more limited, with certain key areas less thoroughly addressed compared to the CN Knowledge Base. Gemini provides extensive coverage, touching on emerging topics such as real-time carbon monitoring, international collaboration, and the role of blockchain technology in carbon management. While Gemini demonstrates innovation and a diverse technological scope, some content, especially regarding advanced concepts like blockchain, may be too technical for the average user, presenting challenges in terms of comprehension.

(2) Technical Depth: The CN Knowledge Base offers substantial technical depth, particularly in core areas like carbon capture, energy optimization, and carbon emission forecasting. While it does not delve into the granular implementation details of each technology, it aligns closely with the practical requirements of the carbon neutrality field. It provides actionable solutions that are feasible and readily applicable for policymakers and industry professionals. GPT-4 excels in technical depth, especially in areas such as AI and machine learning. It offers thorough discussions on topics like predictive modeling and data analysis, demonstrating a deep understanding of how AI can optimize carbon monitoring and management. However, GPT-4 often lacks detailed operational guidance, and its explanations are sometimes too general, which can make it challenging for non-technical users to apply the concepts in real-world scenarios. Bing AI, on the other hand, takes a more general approach to technical depth, focusing primarily on the high-level applications of AI in the carbon neutrality space. While it offers useful conceptual overviews, it does not provide the detailed technical implementations or solutions necessary for more advanced or technical inquiries. As such, it is better suited for users seeking introductory information but falls short for those looking for specific, actionable technical solutions. Gemini provides a broad and detailed exploration of cutting-edge technologies, such as the integration of AI with blockchain, intelligent sensor networks, and personalized carbon reduction strategies. While these topics are highly innovative, some of the content is highly technical, posing comprehension challenges for users without a technical background. Nevertheless, Gemini offers a comprehensive overview of state-of-the-art solutions in the carbon neutrality space.

(3) Practical Operability: The CN Knowledge Base excels in practical operability, offering a variety of real-world application scenarios and well-defined implementation pathways. For example, it provides clear instructions for optimizing energy management through AI, enhancing the efficiency of CCS, and improving supply chain management for carbon neutrality. These actionable steps make the knowledge base highly applicable for policymakers and industry professionals seeking concrete guidance on implementing carbon-neutral technologies. GPT-4 also demonstrates strong practical operability, particularly in areas like carbon emission forecasting via AI models and intelligent grid optimization, with relatively clear implementation routes. However, it remains limited in its guidance on policy-level applications and specific business processes. In practice, many of its steps are vague and lack the detailed, actionable instructions necessary for real-world implementation, which could pose a challenge for users seeking to apply these concepts in tangible projects. Bing AI's practical applicability is more limited. While it describes several relevant use cases, it does not provide enough operational details or practical guidance for implementation. It is more suitable for users with a basic understanding of carbon-neutral technologies, but those seeking comprehensive operational instructions may find it insufficient. Gemini, while presenting highly innovative technological applications, has somewhat limited practical operability. Its focus on complex, cutting-edge technologies means that the guidance it offers for implementation, particularly for businesses and policymakers, is less substantial. As a result, it may provide limited value to practitioners seeking clear, actionable steps for applying these technologies in the real world.

(4) Simplicity and Understandability: The CN Knowledge Base strikes a balance between comprehensiveness and clarity, offering answers that are both detailed and succinct. It avoids excessive technical jargon and complex explanations, making the content accessible to a broad audience, including non-specialists, policymakers, and industry practitioners. The language is clear and user friendly, ensuring that even readers with limited technical expertise can easily grasp the key concepts. GPT-4's responses are generally clear but may present challenges for non-expert readers due to their technical nature. While it conveys detailed application insights effectively for those with a technical background, the complexity of the language and explanations may make it difficult for non-experts to fully comprehend the insights, limiting its accessibility for a wider audience. Bing AI excels in simplicity, avoiding unnecessary technical details, which makes it ideal for users who need a quick overview or basic understanding of carbon-neutral technologies. However, this simplicity comes at the cost of depth, meaning Bing AI is more suited for introductory-level information rather than in-depth, actionable guidance. Gemini, while showcasing a wide array of cutting-edge technologies, may be overwhelming for general users due to the volume of information and its highly specialized nature. For those without a technical background, the content can be complex and challenging to understand, limiting its accessibility to a broader audience.

(5) Overall Evaluation: The CN Knowledge Base stands out as the most well-rounded resource, offering a balanced mix of content comprehensiveness, practicality, and simplicity. This makes it an excellent choice for a wide range of users, especially policymakers and industry professionals, who need both in-depth information and actionable guidance. GPT-4 excels in technical depth, particularly in areas like AI and data analysis, providing valuable insights for users with technical expertise. However, its lack of focus on practical applicability and simplicity limits its use for non-specialists, making it better suited for users with a technical background. Bing AI is the simplest and most accessible option, providing easy-to-understand content without delving deeply into technical details. While this makes it an ideal tool for users seeking an introductory overview of carbon-neutral technologies, its lack of depth restricts its utility for those seeking more detailed or specialized information. Gemini, while showcasing innovative technologies and a broad range of application scenarios, tends to be more complex and specialized. It is better suited for professionals and researchers with technical expertise, as the content may be difficult for general users to grasp and apply effectively.

4.3. Performance Evaluation

As shown in Figure 3, a comparative performance analysis of eight models—the CN Knowledge Base, GPT-4, Gemini, Bing, ChatGLM3, Kimi, Doubao, and ERNIE Bot—was conducted. The evaluation employed three key assessment metrics, BLEU Score, BERT Score, and METEOR Score, to analyze the variations in algorithmic mechanisms and training strategies, and to explore how these differences impact model performance. The results reveal that GPT-4, with its extensive pre-training on vast corpora of data and advanced Transformer architecture, achieved high performance, with a BLEU Score of 0.8784 and BERT Score of 0.8404. These scores highlight GPT-4's exceptional text generation accuracy and semantic understanding, which are driven by its broad exposure to general-purpose datasets. This comprehensive training equips GPT-4 with strong generalization abilities and an impressive aptitude for capturing contextual meaning. However, as a general-purpose large model, GPT-4's adaptability to domain-specific topics like carbon neutrality is somewhat limited. This is reflected in its METEOR Score (0.2655), which indicates reduced fluency in text generation when dealing with specialized terminology, and challenges in aligning its outputs with domain-specific content.

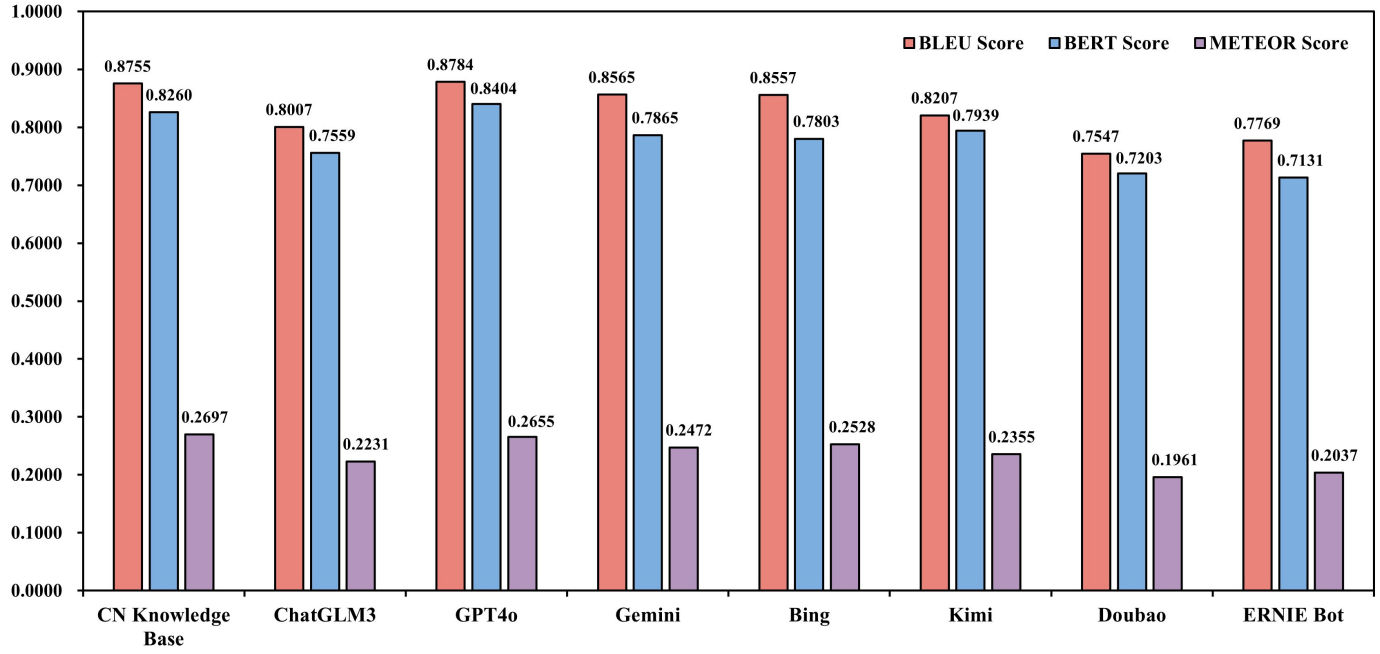


Figure 3. Assessment result.

The CN Knowledge Base demonstrated exceptional performance in this evaluation, particularly in METEOR Score (0.2697), which highlights its strengths in text fluency, expression quality, and word order optimization. Additionally, its BLEU Score (0.8755) and BERT Score (0.8260) closely follow GPT-4o, indicating that its generated text also excels in terms of accuracy and semantic consistency. The model's key strengths are primarily reflected in the following aspects: First, training on domain-specific corpora equips the model to precisely capture the terminology, concepts, and expressions of the carbon-neutral sector. This, coupled with its reliance on high-quality carbon-neutral literature and data reports, significantly enhances its expertise. Second, the CN Knowledge Base likely incorporates RAG, which improves the professionalism and coherence of its responses by matching semantic vectors and integrating high-quality text fragments. Finally, the structuring and optimization of the generated text further improve its coherence and organization, ensuring that the responses align closely with standard answers in terms of word order, synonym matching, and semantic fluency, thus contributing to a higher METEOR Score. Overall, the model's high degree of domain-specific adaptability provides it with a distinct advantage in carbon-neutrality-related tasks.

Gemini and Bing displayed consistent performance in this evaluation, with BLEU Scores of 0.8565 for Gemini and 0.8557 for Bing and BERT Scores of 0.7865 for Gemini and 0.7803 for Bing. These results reflect their strong generalization abilities in lexical matching and semantic consistency. This can be attributed to their pre-training on large-scale, general-purpose datasets, as well as the optimization of their decoding mechanisms. However, their METEOR Scores—0.2472 for Gemini and 0.2528 for Bing—are relatively low, suggesting that further improvements are needed in terms of text fluency, structured expression, and the adaptation of their models to domain-specific terminology, particularly in the carbon neutrality field.

ChatGLM3 and Kimi demonstrated moderate performance in this evaluation. Their BLEU Scores—0.8007 for ChatGLM3 and 0.8207 for Kimi—indicate a basic level of lexical matching. However, their relatively lower BERT Scores and METEOR Scores reflect limitations in semantic understanding and expression quality. These shortcomings are likely due to their focus on general-purpose tasks during pre-training, without deep optimization for domain-specific corpora. As a result, both models underperformed on specialized topics

such as carbon neutrality. In particular, they struggled with generating coherent, logically rigorous professional text, highlighting the need for better adaptation to the specialized terminology and requirements of the field.

Doubao and ERNIE Bot exhibited poorer performance across three evaluation metrics: BLEU (Doubao: 0.7547, ERNIE Bot: 0.7769), BERT (Doubao: 0.7203, ERNIE Bot: 0.7131), and METEOR (Doubao: 0.1961, ERNIE Bot: 0.2037). This underperformance can be attributed to the limited scale and quality of their training data, along with insufficient exposure to domain-specific knowledge in the carbon neutrality field. Algorithmically, both models face challenges in context-based semantic modeling and text generation optimization, which impedes their ability to generate accurate, professional, and fluent responses. These limitations are reflected in the models' overall inability to produce high-quality answers in specialized areas like carbon neutrality.

In summary, the CN Knowledge Base stands out for its high-quality data collection, structured knowledge optimization, and domain-specific adaptation. It demonstrates strong alignment with standard answers, particularly excelling in professionalism and expression quality, as reflected in its METEOR Score. This highlights the model's expertise in optimizing language structure and enhancing text fluency. In contrast, GPT-4o shows notable strengths in semantic understanding and text generation accuracy. However, it falls short of the CN Knowledge Base in its ability to adapt to domain-specific terminology and maintain high-quality expression in specialized fields. Gemini and Bing, while performing steadily across general tasks, exhibit a limited capacity to adapt to specialized domains, making them less effective than domain-specific models for topics like carbon neutrality. Finally, the performance of ChatGLM3, Kimi, Doubao, and ERNIE Bot underscores their limitations in domain-specific knowledge accumulation and algorithmic optimization, which affect their ability to generate accurate, fluent, and professional responses in specialized contexts.

From an algorithmic perspective, the success of the CN Knowledge Base highlights the distinct advantages of domain-specific models. Through targeted data training and structural optimization, these models are able to achieve precise adaptation for language understanding, semantic consistency, and expression quality within specialized domain tasks. In contrast, general models like GPT-4o, while highly effective in generalization, still require further refinement to achieve optimal performance in specialized fields. The exceptional performance of the CN Knowledge Base not only demonstrates the feasibility of deeply integrating domain-specific knowledge with generative algorithms, but also provides strong support for the implementation of carbon neutrality policies and technological applications. This underscores the irreplaceable and forward-thinking value of domain-specific models in real-world applications.

5. Discussion

This study investigates the initial application of the ChatGLM3-based large pre-trained language model in building a specialized CN Knowledge Base. It introduces a RAG approach, which integrates an external knowledge base to enhance the model's responses. The study develops a carbon neutrality DSQA system using the ChatGLM3-6B model and the Langchain open-source framework. This system significantly improves the automatic answering capability for carbon-neutrality-related queries, demonstrating the potential and effectiveness of domain-specific LLMs in addressing challenges in the field of carbon neutrality. Additionally, the study conducts a quantitative assessment and comparative evaluation of several mainstream models, including GPT-4o, Gemini, Bing, Kimi, Doubao, and ERNIE Bot, analyzing their performance characteristics and identifying the underlying factors contributing to performance differences.

5.1. System Advantages

The CN Knowledge Base developed in this study is built upon the ChatGLM3 large pre-trained language model, which has been specifically optimized and fine-tuned for the carbon neutrality domain. This adaptation demonstrates significant advantages in QA tasks related to carbon neutrality. According to the evaluation metrics, the knowledge base excels with an METEOR Score of 0.2697, surpassing other models and showcasing a high level of language fluency, word order optimization, and structural coherence in its generated responses. Furthermore, its BLEU Score (0.8755) and BERT Score (0.8260) are comparable to the best-performing models like GPT-4o, highlighting its competitive performance in terms of accuracy and semantic consistency. These results indicate that the ChatGLM3-based large language model demonstrates strong task adaptability and high-quality generation in the carbon neutrality field, effectively addressing the complex knowledge needs of tasks such as carbon emission monitoring, policy interpretation, and technical optimization.

The underlying mechanisms driving its advantages are reflected in three main aspects: First, the CN Knowledge Base is trained using a wealth of authoritative carbon neutrality literature, policy documents, and academic reports, ensuring the model's deep understanding of domain-specific terminology, concepts, and logic. This enables the generation of more accurate and targeted answers, significantly improving the output quality. Second, the study integrates RAG technology, which further enhances the model's ability to integrate and generate knowledge within the carbon neutrality domain. Through vectorized semantic matching, the model retrieves high-quality and relevant text segments, strengthening the accuracy and relevance of the answers. Compared to general-purpose models, this structured data processing and knowledge retrieval mechanism substantially enhances the expertise and applicability of the generated responses. Finally, the superior METEOR Score reflects the model's optimization capabilities in terms of fluency, word order organization, and linguistic naturalness. Due to the structured processing and optimization of domain-specific data, the model is better able to adhere to the expression conventions specific to the carbon neutrality field, effectively meeting practical application needs.

This study further highlights the advantages of the CN Knowledge Base in domain-specific tasks through a comparative evaluation with leading models such as GPT-4o, Gemini, and Bing. GPT-4o, with its extensive pre-training and strong generalization capabilities, excels in BLEU Score (0.8784) and BERT Score (0.8404), demonstrating its superiority in text generation accuracy and semantic understanding. However, since GPT-4o has not been deeply adapted to domain-specific data, it slightly underperforms compared to the CN Knowledge Base in terms of language structure optimization and terminology expression, leading to a lower METEOR Score (0.2655). Gemini and Bing show consistent performance in the evaluation, reflecting their strong capabilities in general tasks. However, their performance in the carbon neutrality domain is less effective due to insufficient domain-specific optimization. This lack of adaptation affects the precision of their language expression and terminology matching, particularly when compared to models specifically tailored for the domain. In contrast, ChatGLM3 and Kimi demonstrate moderate performance. While they achieve reasonable BLEU Scores, they still exhibit some deficiencies in semantic consistency and fluency, highlighting the challenges faced by general-purpose models in adapting to specialized tasks. Doubao and ERNIE Bot show the weakest performance, primarily due to the limited scale of their training data and inadequate domain adaptation, underscoring the critical importance of high-quality domain-specific data and targeted task-specific optimization for achieving optimal model performance.

This study demonstrates the strong applicability and scalability of the ChatGLM3-based large language model in constructing a CN Knowledge Base, offering reliable,

intelligent QA support for tasks such as carbon neutrality policy evaluation, technical path optimization, and carbon footprint monitoring. The research highlights the significant advantages of domain-specific models in addressing targeted tasks, particularly in their ability to adapt professional knowledge, ensure expression fluency, and achieve content precision—outperforming mainstream general-purpose models. This exploration paves a new path for the digitalization and intelligent transformation of the carbon neutrality field, contributing to the advancement of global carbon neutrality goals.

5.2. Limitations and Future Improvements

Although this study has yielded promising results, there are several opportunities for further optimization. First, the carbon neutrality corpus could be expanded to include additional technical solutions, policy papers, and case studies, which would enhance the knowledge base's comprehensiveness and its ability to update dynamically. To maintain the knowledge base's timeliness, future work should prioritize the development of dynamic update mechanisms, particularly in light of the rapid advancements in technology, policy, and practice within the field of carbon neutrality. The integration of automated data scraping and updating systems, complemented by expert feedback and validation, will ensure that the knowledge base remains aligned with the latest developments and emerging trends in the field. Second, future work could explore a hybrid approach by integrating the ChatGLM3 domain-specific model with general-purpose models like GPT-4o. This would leverage their combined strengths in domain adaptability and generalization, improving the accuracy and relevance of generated responses. Additionally, incorporating multimodal data (e.g., images and geographic information system data) and reinforcement learning mechanisms could further enhance the model's performance in complex tasks such as carbon emission monitoring and intelligent decision-making.

In terms of the evaluation methodology, this study utilized several natural language processing metrics, including BLEU, BERT Score, and METEOR, for quantitative analysis. However, expert evaluation was not incorporated for the following reasons: First, the scarcity of domain experts in carbon neutrality, coupled with the large volume of data, made expert involvement in the evaluation impractical. Second, expert assessments typically require substantial time and resources, posing challenges for their application in large-scale datasets. As a result, this study employed multiple scoring methods to strike a balance between objectivity and comprehensiveness, ensuring that the evaluation results accurately reflect the system's performance and effectiveness.

In conclusion, this study, through the preliminary development of a CN Knowledge Base based on ChatGLM3, demonstrates the distinct advantages of domain-specific LLMs in specialized tasks. It also highlights the complementary relationship between domain-specific models and general-purpose models like GPT-4o, offering novel intelligent support for carbon neutrality policy implementation and technological development. This research not only showcases the potential of large language models in creating specialized knowledge bases but also provides both theoretical insights and practical pathways for optimizing and expanding the applications of future intelligent QA systems.

6. Conclusions

This study explores the preliminary application of the ChatGLM3-based large language model in building a CN Knowledge Base, aiming to assess the effectiveness and potential of domain-specific large language models for carbon-neutrality-related QA tasks. The evaluation results show that the ChatGLM3-based CN Knowledge Base achieves exceptional performance, with a BLEU Score of 0.8755, a BERT Score of 0.8260, and an METEOR Score of 0.2697. Its METEOR Score in particular surpasses that of other models such as

GPT-4o (0.2655), Gemini (0.2472), and Bing (0.2528), underscoring its advantages in linguistic fluency, terminology adaptation, and textual expression. Particularly in addressing domain-specific queries and handling carbon-neutrality-related terminology, ChatGLM3, trained on domain-specific corpora and potentially enhanced by the RAG mechanism, exhibits strong adaptability and generates high-quality responses.

In contrast, GPT-4o, with its large-scale, multi-domain data and strong generalization capabilities, excels in BLEU Score (0.8784) and BERT Score (0.8404), particularly in semantic consistency and text generation accuracy. However, despite its strengths in general tasks, GPT-4o has not been specifically optimized for the carbon neutrality domain, resulting in somewhat weaker performance in terminology expression and domain-specific adaptation compared to the CN Knowledge Base. Gemini and Bing show stable performance in general tasks, but, due to insufficient domain adaptation, their language structuring and professional task performance do not reach the level of the CN Knowledge Base. Meanwhile, ChatGLM3, Kimi, Doubao, and ERNIE Bot exhibit comparatively lower performance in carbon-neutrality-related tasks, primarily due to limited domain data coverage and lack of optimization.

The superior performance of the CN Knowledge Base can be attributed to its training on domain-specific corpora and the potential application of the RAG mechanism. By deeply integrating specialized knowledge from the carbon neutrality domain, the model excels in generating responses for tasks such as policy interpretation, technology path optimization, and carbon emission monitoring. These results highlight the unique advantages of domain-specific models in addressing specialized tasks and provide significant practical insights for the development of intelligent QA systems in this field.

In future research, as domain data continue to expand and efforts to integrate and optimize both general-purpose and domain-specific models progress, the applicability and performance of the CN Knowledge Base are expected to significantly improve. Moreover, the incorporation of multimodal data processing technologies, such as image and geospatial data, will be pivotal in tasks like advanced carbon emission monitoring and intelligent decision-making support. Although this study utilized multiple natural language processing metrics for model evaluation, expert evaluation was not included due to the large data volume and the impracticality of involving experts. Instead, a variety of scoring methods were used for quantitative analysis. Despite these limitations, this research provides crucial support for the informatization and intelligent development of carbon neutrality and offers valuable insights for constructing intelligent knowledge bases in other fields. Ultimately, it contributes technological support to advancing the global carbon neutrality goal.

Author Contributions: L.L.: writing—original draft, visualization, validation, software, methodology, formal analysis, data curation, conceptualization. Y.Z. (Yongzhang Zhou): supervision, resources, project administration, investigation, funding acquisition. J.M.: data curation, investigation, methodology, conceptualization. Y.Z. (Yuqing Zhang): data curation, investigation, methodology, conceptualization. L.H.: writing—original draft, software, methodology, conceptualization, data curation, formal analysis. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Key Research and Development Program of China (2022YFF0801201) and supported by the National Natural Science Foundation of China (U1911202) and the Guangdong Key Areas Research and Development Project (2020B1111370001).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data presented in this study are available upon request from the corresponding author.

Acknowledgments: We would like to thank Sun Yat-sen University for providing research equipment and resources for this study.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Wang, P.; Yang, Y.; Xue, D.; Ren, L.; Tang, J.; Leung, L.R.; Liao, H. Aerosols overtake greenhouse gases causing a warmer climate and more weather extremes toward carbon neutrality. *Nat. Commun.* **2023**, *14*, 7257. [CrossRef] [PubMed]
2. Masson-Delmotte, V.; Zhai, P.; Pirani, A.; Connors, S.L.; Péan, C.; Berger, S.; Caud, N.; Chen, Y.; Goldfarb, L.; Gomis, M.I.; et al. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*; Cambridge University Press: Cambridge, UK, 2021; Volume 2, p. 2391. [CrossRef]
3. Toromade, A.S.; Soyombo, D.A.; Kupa, E.; Ijomah, T.I. Reviewing the impact of climate change on global food security: Challenges and solutions. *Int. J. Appl. Res. Soc. Sci.* **2024**, *6*, 1403–1416. [CrossRef]
4. Zhou, T.; Chen, X.; Zhang, W.; Zhang, L. Climate change and carbon neutrality. *Chin. J. Nat.* **2024**, *46*, 1–11. [CrossRef]
5. Kabeyi, M.J.B.; Olanrewaju, O.A. Sustainable energy transition for renewable and low carbon grid electricity generation and supply. *Front. Energy Res.* **2022**, *9*, 743114. [CrossRef]
6. Esiri, A.E.; Kwakye, J.M.; Ekechukwu, D.E.; Ogundipe, O.B.; Ikevuje, A.H. Leveraging regional resources to address regional energy challenges in the transition to a low-carbon future. *Open Access Res. J. Multidiscip. Stud.* **2024**, *8*, 105–114. [CrossRef]
7. Huang, J.; Hu, B.; Meng, J.; Meng, T.; Liu, W.; Guan, Y.; Zhang, X. Highly efficient sustainable strategies toward carbon-neutral energy production. *Energy Environ. Sci.* **2024**, *17*, 1007–1045. [CrossRef]
8. Su, C.W.; Liu, F.; Stefea, P.; Umar, M. Does technology innovation help to achieve carbon neutrality? *Econ. Anal. Policy* **2023**, *78*, 1–14. [CrossRef]
9. Zhu, H.; Jiang, M.; Zhang, D.; Goh, H.H.; Wang, S.; Mao, D.; Wu, T. Carbon neutrality pathways exploration—A state-of-the-art study: Key technological advancements, current challenges, and potential opportunities. *Sustain. Energy Technol. Assess.* **2023**, *60*, 103489. [CrossRef]
10. Raza, S.; Ghasali, E.; Raza, M.; Chen, C.; Li, B.; Orooji, Y.; Erk, N. Advances in technology and utilization of natural resources for achieving carbon neutrality and a sustainable solution to neutral environment. *Environ. Res.* **2023**, *220*, 115135. [CrossRef]
11. Guntuka, L.; Mukandwal, P.S.; Aktas, E.; Paluvadi, V.S.K. From carbon-neutral to climate-neutral supply chains: A multidisciplinary review and research agenda. *Int. J. Logist. Manag.* **2024**, *35*, 916–942. [CrossRef]
12. Sun, L.L.; Cui, H.J.; Ge, Q.S. Will China achieve its 2060 carbon neutral commitment from the provincial perspective? *Adv. Clim. Chang. Res.* **2022**, *13*, 169–178. [CrossRef]
13. Alola, A.A.; Adebayo, T.S. The potency of resource efficiency and environmental technologies in carbon neutrality target for Finland. *J. Clean. Prod.* **2023**, *389*, 136127. [CrossRef]
14. Wen, Y.; Zheng, L.; Leng, D.; Dai, C.; Lu, J.; Zhang, Z.; Bo, X. Deep learning-based multi-omics data integration methods for biomedical applications. *Adv. Intell. Syst.* **2023**, *5*, 2200247. [CrossRef]
15. Horvath, S.; Laskaridis, S.; Almeida, M.; Leontiadis, I.; Venieris, S.; Lane, N. Fjord: Fair and accurate federated learning under heterogeneous targets with ordered dropout. *Adv. Neural Inf. Process. Syst.* **2021**, *34*, 12876–12889. [CrossRef]
16. Zhu, W.; Gui, R.; Guo, R. Unveiling the nexus and promoting integration of diverse factors: Prospects of big data-driven artificial intelligence technology in achieving carbon neutrality in Chongming District. *Water Energy Nexus* **2023**, *6*, 112–121. [CrossRef]
17. Liu, Y.; He, H.; Han, T.; Zhang, X.; Liu, M.; Tian, J.; Ge, B. Understanding LLMS: A comprehensive overview from training to inference. *arXiv* **2024**, arXiv:2401.02038. [CrossRef]
18. Vaghefi, S.A.; Huggel, C.; Muccione, V.; Khashehchi, H.; Leippold, M. Deep climate change: A dataset and adaptive domain pre-trained language models for climate change-related tasks. In Proceedings of the NeurIPS 2022 Workshop on Tackling Climate Change with Machine Learning, Virtual, 9 December 2022.
19. Bulian, J.; Schäfer, M.S.; Amini, A.; Lam, H.; Ciaramita, M.; Gaiarin, B.; Huebscher, M.C.; Buck, C.; Mede, N.G.; Leippold, M.; et al. Assessing large language models on climate information. *arXiv* **2023**, arXiv:2310.02932. [CrossRef]
20. Ren, S.; Tomlinson, B.; Black, R.W.; Torrance, A.W. Reconciling the contrasting narratives on the environmental impact of large language models. *Sci. Rep.* **2024**, *14*, 26310. [CrossRef]
21. Lu, R.-S.; Lin, C.-C.; Tsao, H.-Y. Empowering large language models to leverage domain-specific knowledge in e-learning. *Appl. Sci.* **2024**, *14*, 5264. [CrossRef]
22. Augenstein, I.; Baldwin, T.; Cha, M.; Chakraborty, T.; Ciampaglia, G.L.; Corney, D.; Zagni, G. Factuality challenges in the era of large language models and opportunities for fact-checking. *Nat. Mach. Intell.* **2024**, *6*, 852–863. [CrossRef]
23. Farquhar, S.; Kossen, J.; Kuhn, L.; Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature* **2024**, *630*, 625–630. [CrossRef]

24. Izacard, G.; Grave, E. Distilling knowledge from reader to retriever for question answering. *arXiv* **2020**, arXiv:2012.04584. [CrossRef]
25. Etezadi, R.; Shamsfard, M. The state of the art in open domain complex question answering: A survey. *Appl. Intell.* **2023**, *53*, 4124–4144. [CrossRef]
26. Kaddour, J.; Harris, J.; Mozes, M.; Bradley, H.; Raileanu, R.; McHardy, R. Challenges and applications of large language models. *arXiv* **2023**, arXiv:2307.10169. [CrossRef]
27. Ding, N.; Qin, Y.; Yang, G.; Wei, F.; Yang, Z.; Su, Y.; Sun, M. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nat. Mach. Intell.* **2023**, *5*, 220–235. [CrossRef]
28. Vulpescu, E.; Beldean, M. Optimized fine-tuning of large language model for better topic categorization with limited data. *arXiv* **2024**. [CrossRef]
29. Lee, U.; Jung, H.; Jeon, Y.; Sohn, Y.; Hwang, W.; Moon, J.; Kim, H. Few-shot is enough: Exploring ChatGPT prompt engineering method for automatic question generation in English education. *Educ. Inf. Technol.* **2024**, *29*, 11483–11515. [CrossRef]
30. Giray, L. Prompt engineering with ChatGPT: A guide for academic writers. *Ann. Biomed. Eng.* **2023**, *51*, 2629–2633. [CrossRef]
31. Ji, Z.; Lee, N.; Frieske, R.; Yu, T.; Su, D.; Xu, Y.; Ishii, E.; Bang, Y.J.; Madotto, A.; Fung, P. Survey of hallucination in natural language generation. *ACM Comput. Surv.* **2023**, *55*, 1–38. [CrossRef]
32. Fan, W.; Ding, Y.; Ning, L.; Wang, S.; Li, H.; Yin, D.; Li, Q. A survey on RAG meeting LLMs: Towards retrieval-augmented large language models. In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Barcelona, Spain, 25–29 August 2024; pp. 6491–6501. [CrossRef]
33. Arslan, M.; Ghanem, H.; Munawar, S.; Cruz, C. A survey on RAG with LLMs. *Procedia Comput. Sci.* **2024**, *246*, 3781–3790. [CrossRef]
34. Li, D.; Li, Z.; Yang, Y. Knowledge graph-enhanced large language model for domain-specific question answering systems. *TechRxiv* **2024**. [CrossRef]
35. Omar, R.; Dhall, I.; Kalnis, P.; Mansour, E. A universal question-answering platform for knowledge graphs. *Proc. ACM Manag. Data* **2023**, *1*, 1–25. [CrossRef]
36. Ma, R.; Liu, Y.; Ma, Z. f-KGQA: A fuzzy question answering system for knowledge graphs. *Fuzzy Sets Syst.* **2025**, *498*, 109117. [CrossRef]
37. Sovrano, F.; Palmirani, M.; Vitali, F. Legal knowledge extraction for knowledge graph-based question-answering. In *Legal Knowledge and Information Systems*; IOS Press: Amsterdam, The Netherlands, 2020; pp. 143–153. [CrossRef]
38. Bulla, M.; Hillebrand, L.P.; Lübbering, M.; Sifa, R. Knowledge graph-based question answering system for financial securities. In *Deutsche Jahrestagung für Künstliche Intelligenz*; Springer International Publishing: Cham, Switzerland, 2021. [CrossRef]
39. Liu, C.; Ji, X.; Wang, Y.Y. Chinese mineral question and answering system based on knowledge graph. *Expert Syst. Appl.* **2023**, *231*, 120841.1–120841.8. [CrossRef]
40. Xiong, H.; Wang, S.; Tang, M.; Wang, L.; Lin, X. Knowledge graph question answering with semantic-oriented fusion model. *Knowl.-Based Syst.* **2021**, *221*, 106954. [CrossRef]
41. Jia, Z.; Pramanik, S.; Saha Roy, R.; Weikum, G. Complex temporal question answering on knowledge graphs. In Proceedings of the 30th ACM International Conference on Information & Knowledge Management, Virtual Event, 1–5 November 2021; pp. 792–802.
42. Yang, Z.; Wang, Y.; Gan, J.; Li, H.; Lei, N. Design and research of intelligent question-answering (Q&A) system based on high school course knowledge graph. *Mob. Netw. Appl.* **2021**, *26*, 1884–1890. [CrossRef]
43. Yani, M.; Krisnadhi, A.A. Challenges, techniques, and trends of simple knowledge graph question answering: A survey. *Information* **2021**, *12*, 271. [CrossRef]
44. Wang, F.; Shi, D.; Aguilar, J.; Cui, X.; Jiang, J.; Shen, L.; Li, M. LLM-KGMQA: Large Language Model-Augmented Multi-Hop Question-Answering System Based on Knowledge Graph in Medical Field. *PREPRINT*, (Version 1); 4 August 2024. Available online: https://www.researchsquare.com/article/rs-4721418/v1 (accessed on 20 December 2024).
45. Soman, K.; Rose, P.W.; Morris, J.H.; Akbas, R.E.; Smith, B.; Peetoom, B.; Villouta-Reyes, C.; Cerono, G.; Shi, Y.; Baranzini, S.E.; et al. Biomedical knowledge graph-optimized prompt generation for large language models. *Bioinformatics* **2024**, *40*, btae560. [CrossRef] [PubMed]
46. Oduro-Afriyie, J. Improving Personal Health Question Answering Through Knowledge Graph Enhancement. Doctoral Dissertation, University of Idaho, Moscow, ID, USA, 2024. [CrossRef]
47. Bui, T.; Tran, O.; Nguyen, P.; Ho, B.; Nguyen, L.; Bui, T.; Quan, T. Cross-data knowledge graph construction for LLM-enabled educational question-answering system: A case study at HCMUT. In Proceedings of the 1st ACM Workshop on AI-Powered Q&A Systems for Multimedia, Phuket, Thailand, 10 June 2024; pp. 36–43. [CrossRef]
48. Zeng, A.; Xu, B.; Wang, B.; Zhang, C.; Yin, D.; Zhang, D.; Rojas, D.; Feng, G.; Zhao, H.; Wang, Z.; et al. ChatGLM: A family of large language models from GLM-130B to GLM-4 all tools. *arXiv* **2024**, arXiv:2406.12793. [CrossRef]
49. Du, Z.; Qian, Y.; Liu, X.; Ding, M.; Qiu, J.; Yang, Z.; Tang, J. GLM: General language model pre-training with autoregressive blank infilling. *arXiv* **2021**, arXiv:2103.10360. [CrossRef]

50. Topsakal, O.; Akinci, T.C. Creating large language model applications utilizing LangChain: A primer on developing LLM apps fast. In Proceedings of the International Conference on Applied Engineering and Natural Sciences, Konya, Turkey, 10–12 July 2023; Volume 1, pp. 1050–1056. [CrossRef]
51. Wang, Z.; Liu, J.; Zhang, S.; Yang, Y. Poisoned Langchain: Jailbreak LLMs by Langchain. *arXiv* **2024**, arXiv:2406.18122. [CrossRef]
52. Jeong, S.; Baek, J.; Cho, S.; Hwang, S.J.; Park, J.C. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. *arXiv* **2024**, arXiv:2403.14403. [CrossRef]
53. Massey, P.A.; Montgomery, C.; Zhang, A.S. Comparison of ChatGPT-3.5, ChatGPT-4, and Orthopaedic Resident Performance on Orthopaedic Assessment Examinations. *JAAOS-J. Am. Acad. Orthop. Surg.* **2023**, *31*, 1173–1179. [CrossRef]
54. Team, G.; Anil, R.; Borgeaud, S.; Alayrac, J.B.; Yu, J.; Soricut, R.; Blanco, L. Gemini: A Family of Highly Capable Multimodal Models. *arXiv* **2023**, arXiv:2312.11805. [CrossRef]
55. Crawford, J.; Cowling, M.; Ashton-Hay, S.; Kelder, J.A.; Middleton, R. Artificial Intelligence and Authorship Editor Policy: ChatGPT, Bard Bing AI, and Beyond. *J. Univ. Teach. Learn. Pract.* **2023**, *20*, 1–11. [CrossRef]
56. Chen, J.; Li, S.; Huang, Q.; Yan, S.; Xie, Z.; Lu, Y. Application of Kimi Intelligent Assistant in the Teaching of Water Pollution Control Engineering Course. *Int. J. Educ. Hum. Sci.* **2024**, *13*, 39–43. [CrossRef]
57. Yu, C.; Shi, Y. The Age of AI-Enabled Education: The Phenomenon of "Zero Translation" in the Translation of Political Documents from the Perspective of Social Science. In Proceedings of the 2024 International Symposium on Artificial Intelligence for Education, Xi'an, China, 6–8 September 2024; pp. 155–159. [CrossRef]
58. Peng, Q.; Yuan, H. Study on Generating Language Experiment Reading Materials through ERNIE Bot. In Proceedings of the 2024 International Symposium on Artificial Intelligence for Education, Xi'an, China, 6–8 September 2024; pp. 535–539.
59. Reiter, E. A Structured Review of the Validity of BLEU. *Comput. Linguist.* **2018**, *44*, 393–401. [CrossRef]
60. Koroteev, M.V. BERT: A Review of Applications in Natural Language Processing and Understanding. *arXiv* **2021**, arXiv:2103.11943. [CrossRef]
61. Behm, P.; Benoit, P.; Faivre, A.; Meynadier, J.M. METEOR: A Successful Application of B in a Large Project. In Proceedings of the International Symposium on Formal Methods; Springer: Berlin/Heidelberg, Germany, 1999; pp. 369–387. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.